Geocomputation & Machine Learning course

Quick recap

The meeting focused on a technical lecture and practical demonstration of machine learning models, specifically Support Vector Machines (SVM) and the Perceptron. Raffaele and Giuseppe opened with a discussion on overfitting and the use of static versus dynamic features in random forest models, suggesting that training a second model on residuals could be a viable approach. Raffaele then reviewed SVM for classification and regression (SVR), explaining the optimal hyperplane, support vectors, hinge loss, and the kernel trick for non-linearly separable data. He also covered the mathematical foundations of optimization, including gradients, loss functions like MSE and MAE, and the role of regularization (L1 and L2) in improving model robustness. The session transitioned to the Perceptron, where Raffaele explained its structure, activation functions like sigmoid, tanh, and ReLU, and the backpropagation process. A hands-on tutorial using PyTorch demonstrated the implementation of a Perceptron for tree height prediction, showing the impact of activation functions and data scaling on model performance. The conversation ended with a brief logistical discussion between Giuseppe and Annabell regarding travel and accommodation for an upcoming event.

Next steps

Annabell

- Contact the hostel directly to confirm the booking and late arrival details using the code provided by Giuseppe.
- Aim to arrive before Sunday afternoon for the icebreaker event, ideally flying on Saturday or Sunday morning.

Johannes

- Try pruning out features that have a single constant value across the dataset to see if it reduces overfitting in the random forest model.

- Experiment with a two-step modeling approach: first run a random forest with static features, then use the residuals as the response variable for a second random forest with dynamic features.

Raffaele

- Update the PowerPoint slide to explicitly label the SVR loss function as MSE loss to avoid confusion with MAE.

Summary

Support Vector Machines Lecture

Raffaele led a lecture on support vector machines (SVMs), covering both classification and regression approaches. The discussion included concepts like hyperplane optimization, support vectors, and the kernel trick for handling non-linearly separable data. Students asked questions about overfitting in random forest models, particularly regarding static versus dynamic features, which Giuseppe and Raffaele addressed through suggestions about using residual-based modeling approaches. The lecture concluded with interactive questions about SVM concepts, where students discussed the differences between classification and regression approaches, with Annabell providing feedback about clarifying the distinction between MAE and MSE loss functions in the presentation slides.

Derivatives Quiz Discussion

Raffaele paused the presentation to ask the group to complete a quiz on derivatives before continuing. The discussion focused on identifying points on a curve where the derivative is zero, with Raffaele providing a link to a poll for participants to answer. The meeting was temporarily paused with plans to resume later at 10.

Machine Learning Loss Function Optimization

Raffaele discussed the process of minimizing a loss function in machine learning, particularly focusing on how to find the minimum of a function like $f(x) = x^2$. He explained that while finding the derivative and setting it to zero identifies critical points, additional analysis using the second derivative is needed to determine whether these points represent minima, maxima,

or inflection points. Raffaele then outlined the linear regression process, including how to split data into training and testing sets and use the mean squared error (MSE) as a performance metric. The discussion concluded with Raffaele noting that while linear regression can solve problems with linearly distributed data, it cannot handle more complex problems like the XOR logic function.

XOR Logic and Linear Regression

Raffaele explained the XOR logic function and its challenges with linear separation, then discussed linear regression optimization using the mean square error loss function. He described the gradient-based approach to finding optimal weights and presented the solution in matrix form, explaining that for large datasets gradient descent becomes necessary despite the existence of a closed formula. The discussion ended with Raffaele beginning to address a philosophical question about feature selection using tree height and other features.

Feature Importance and Regularization Techniques

Raffaele explained how model training can reveal that certain features contribute nothing to predictions, using an example where feature X2 had a weight of zero despite being semantically important. He demonstrated how sensor noise in production could affect predictions when the model relies solely on a single feature, leading to performance degradation. Raffaele introduced regularization techniques, including L2 normalization and Lasso regression, as solutions to make models more robust to noisy data by allowing them to utilize all features rather than relying solely on the most important one.

Machine Learning Regularization Concepts

Raffaele explained the concept of regularization in machine learning models, emphasizing its importance in making models more reliable and robust to noise in data. He discussed how regularization helps create a smoother loss space, preventing models from getting stuck in local minima. Raffaele also introduced the perceptron model and explained how activation functions like sigmoid and ReLU are used to transform outputs to appropriate ranges, providing examples such as using ReLU to ensure tree height predictions are non-negative.

Activation Functions and Optimization

Raffaele explained the differences between activation functions and loss functions, highlighting issues with computing derivatives at zero and introducing solutions like Leaky ReLU. He then discussed gradient descent optimization, explaining how learning rate and momentum parameters affect the optimization process and the concept of backpropagation. The session concluded with a question-and-answer segment where students answered questions about gradients in machine learning and perceptrons.

Perceptron Tutorial with Torch

Raffaele conducted a tutorial on perceptrons, covering theoretical concepts and their implementation using the Torch framework. He explained the correct answers to questions about perceptron weight updates and activation functions, emphasizing that the main limitation of single-layer perceptrons is their inability to handle non-linear problems. Raffaele then outlined how to define a perceptron in Torch, including the initializer and forward methods, and demonstrated creating and fitting a perceptron model to predict tree height using random data points.

Perceptron Model Training Demonstrations

Raffaele demonstrated different approaches to training a perceptron model, showing how the model's performance changed when using ReLU activation functions and when applying it to real-world data. The model achieved an R-squared value of 0.98 without constraints but performed worse when constrained by ReLU, highlighting the trade-offs between model flexibility and output constraints. When applying the model to latitude and longitude features from a three-height dataset, the group discussed potential reasons for underperformance, including the possibility of underfitting due to insufficient training epochs or using inappropriate features.